



Embodied AI

Lecture 1: Introduction

Junwei Liang
AI Thrust
HKUST (Guangzhou)

Logistics: Part 1

- Course website

- <https://hkust-aiaa4220.github.io/fall2026/lecs.html>
- You can see the **course schedule, homework links, download slides, policy**, etc.
- The slides will be available before each class

You can download the slides now from the schedule website for today's lecture if you want to take note on it

Embodied AI

AIAA 4220, Fall 2026

Instructors: [Junwei Liang](#)



香港科技大学 (广州)
THE HONG KONG
UNIVERSITY OF SCIENCE AND
TECHNOLOGY (GUANGZHOU)

Home		Schedule
Lecture:	Date and Time: 09:00AM - 11:50AM, Monday Location: Rm 233, W1 Websites: Instructure Zoom: 628-334-1826 PW: 12345 (In-person expected) (Recordings will be available after each class)	
Course Description:	This course aims to provide students with key principles and algorithms to build modern Embodied AI systems. Key topics include machine perception, embodied navigation and manipulation algorithms. Through this course, students will learn and practice the foundational principles, techniques, and tools to build new Embodied AI systems.	
Target Audience/ Prerequisites:	This is a course primarily for junior and senior students.	
Course Work:	Grading	Grading will be based on best 5 of 6 in-class quizzes (25%), one paper presentation (25%), one machine perception homework (25%) and one simulation project (25%). • The in-class quizzes are designed to complement the lectures and making sure you understand the key knowledge about the course. • There are 6 in-class quizzes over the semester. We will take your
	In-Class Quizzes	

We will go over the course policy later today

Your Instructor and TAs This Semester (AIAA 4220)



Junwei Liang
junweiliang@hkust-gz.edu.cn
Course lecturer

- Our lectures are scheduled on Monday every week from 9:00 am. to 11:50 am.
- We will take a break every hour

Your TA:

Yuxin He
yhe469@connect.hkust-gz.edu.cn

Jinghao Huang
jhuang401@connect.hkust-gz.edu.cn

Agenda

- A bit about me...
- What is Embodied AI?
- History of Robotics
- Why study Embodied AI now?
- Course content overview
- Logistics

A bit about me ...



Junwei Liang
junweiliang@hkust-gz.edu.cn
AIAA 4220 course lecturer

- I'm from Guangzhou
- I studied in Beijing (@RUC), Pittsburgh (@CMU) and interned in Mountain View (@Google)
- I moved to Shanghai (@Tencent)
- And now I'm back!



Our Lab is the leading Embodied AI lab in the greater bay area

- Check our lab website for more cool demos: <https://precognition.team/> Or check out our 小红书!

Spatial Intelligence → VLM

Mobility - Navigation + Locomotion

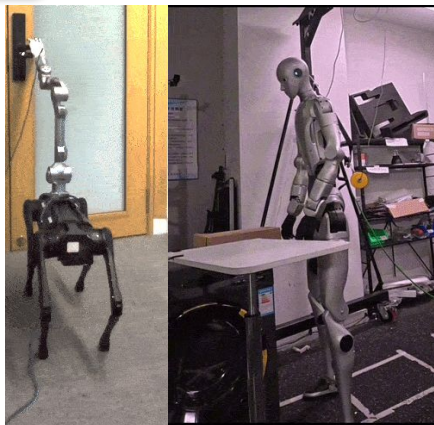
“Action”

Manipulation

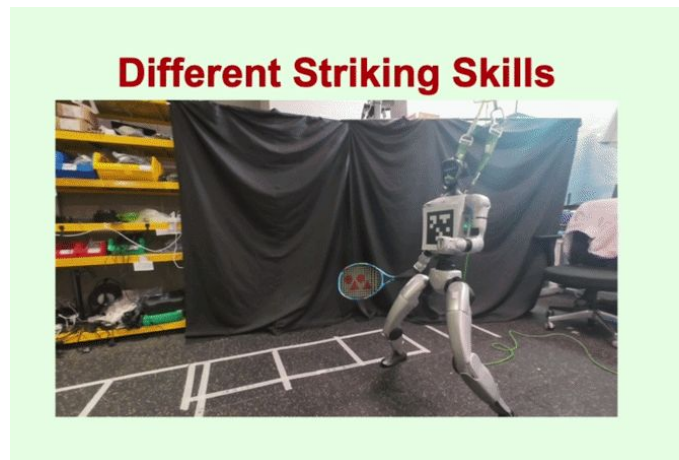
小红书



Agile Locomotion



Manipulation with dex hand



Athletic Intelligence

More cutting-edge research of our lab will be shown today

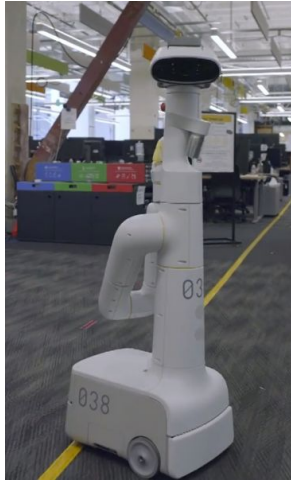
Agenda

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- History of Robotics
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What is Embodied AI?



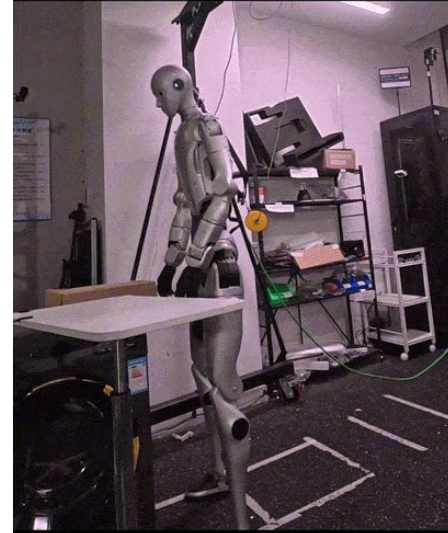
Boston Dynamics



Single-arm Robot



Unitree Robotics



Humanoid

Robots? Two-legged, wheeled, four-legged...

See how they move

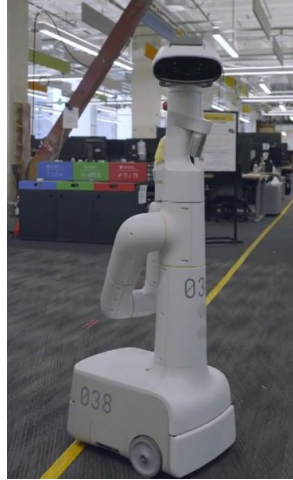
* <https://www.youtube.com/watch?v=Gm6dZ1q06ks>

* https://www.youtube.com/watch?v=-e1_QhJ1EhQ

What is Embodied AI?



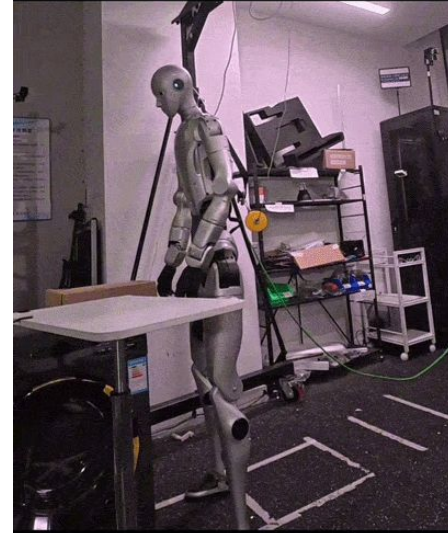
Boston Dynamics



Single-arm Robot



Unitree Robotics



Humanoid

Robots? Any machine that can perform a task automatically!

- Could be based on prespecified rules or through machine learning..

What is Embodied AI?

- Embodied AI is a robot capable of **sensing its environment** and **performing its tasks accordingly** in the real world.
- Key intelligent properties
 - Adaptability (account for noise in the world)
 - Learning from data

Are washing machines considered Embodied AI? No.
There is no environment sensing

Are clearing bots Embodied AI? Maybe. It detects obstacles and clean the house



What is Embodied AI?

- Embodied AI is the intersection of **machine learning and robotics**
 - In Chinese: “具身智能”
- The study of **training optimal solutions using sensory data** for robots to act and achieve goals
- Perception and decision making are the key
- Possible future application
 - General-purpose home robots!



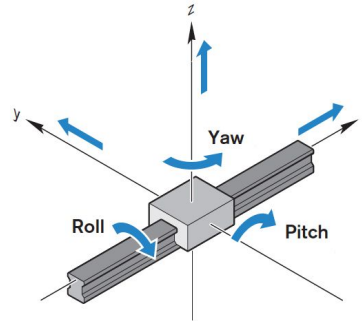
Embodied AI - What for?

- The AI should take over the three “D”s (according to the robotics researcher)
 - The dull
 - Repetitive tasks
 - The dirty
 - The dangerous
 - Warehouse robots
 - Cleaning bots
 - Underwater robots
 - etc.

When Machine Learning is **NOT** Needed

- Industrial Robot

- Populated from 1970s
- Usually a single arm with no mobility
- Move an object with up to 6 DoF
 - Xyz - position
 - pitch, yaw, roll - direction
- Programmed to do repetitive actions (The Dull)
 - Sometime dangerous, lifting heavy stuff
- **Repetitive, non-changing processes**



Define points P1–P5:

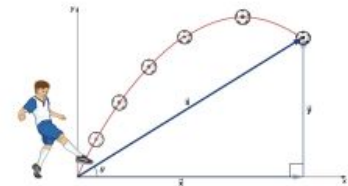
1. Safely above workpiece (defined as P1)
2. 10 cm Above bin A (defined as P2)
3. At position to take part from bin A (defined as P3)
4. 10 cm Above bin B (defined as P4)
5. At position to take part from bin B. (defined as P5)

Define program:

1. Move to P1
2. Move to P2
3. Move to P3
4. Close gripper
5. Move to P2
6. Move to P4
7. Move to P5
8. Open gripper
9. Move to P4
10. Move to P1 and finish

When Machine Learning is **NOT** Needed (for most of the time...)

- Physical processes
 - **Kinematics and inverse kinematics** of different actuators
 - Meaning: how robots (parts) move to a desired position
 - We don't have to "reinvent the wheel". We have good models of physical processes
 - Called **Model-based Control**
 - We can just call their APIs. Although you can still use machine learning to improve
 - With reinforcement learning!
 - Key problems include Sim-to-Real, accounting for the imperfect actuation



When Machine Learning **IS** Needed

- Robots in **unstructured environments**



Structured/known environment



Environment
uncertainty



Object variation

Embodied AI vs. Automation

Special-Purpose Robot Automation

Without machine learning, it is usually called automation, not autonomy



custom-built robots



human expert programming & manual calibration



special-purpose behaviors

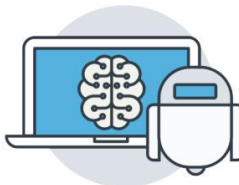
General-Purpose Robot Autonomy



general-purpose robots



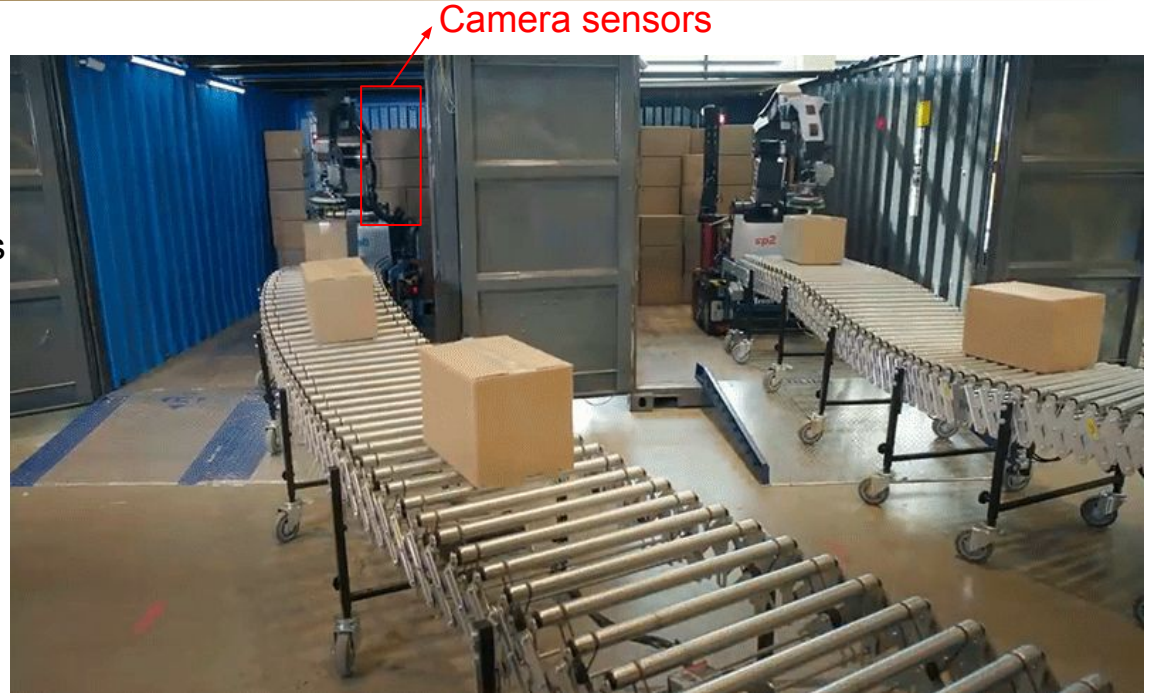
Robot Learning



general-purpose behaviors

Advancement in Unstructured Environment

- (New) Warehouse Robot
 - Boston Dynamics 2022 new product
 - Unloading boxes from trucks to warehouses
 - Adapt to many kinds of boxes
 - Suction gripper



* Learn more about Stretch: <https://www.youtube.com/watch?v=qgGH6Vk8nY8>

Advancement in Unstructured Environment



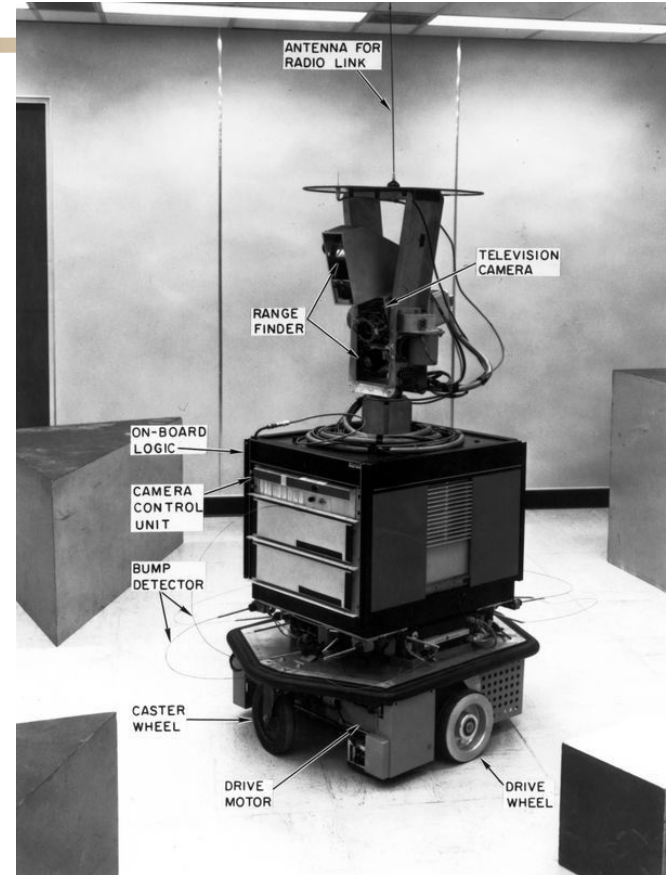
- We are already using some form of Embodied AI products
 - Cleaning robots - key tech: SLAM
 - Pets litter box - pet recognition - infrared and weight sensor
 - Home Security camera - person detection and tracking
 - Self-driving taxi
 - Delivery robots

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History of Robotics

- The first “real” robot
 - 1960s - Shakey the robot - A* algorithm
 - Charles Rosen, Stanford Research Institute



History of Robotics

- 1980s - Robotic Arms are populated in Automation



History of Robotics

- 2000s
 - Cleaning robots - Roomba/iRobot - 2002
 - Simple algorithms
 - Simple bump sensor
 - No mapping of the room
 - Random walk angle changing after bumping into an object or wall



Long exposure photo showing path taken by a Roomba as it cleans



History of Robotics

- 2000s
 - 2004 - 2007 - DARPA Grand Challenge
 - First autonomous driving competition
 - CMU won in 2007, finishing a **96 km** urban course in over **4 hours** (avg. **22.53 km/h**)
 - **So slow it was like a bicycle ride**
 - Already uses modern senses like cameras, lidars and radars*

* <http://www.tartanracing.org/tech.html>

* paper: https://www.ri.cmu.edu/pub_files/2007/4/Tartan_Racing.pdf



Today's Robotics

- 2010s - 2025
 - Consumer-grade drones - DJI
 - Self-driving cars - Waymo, etc.
 - Highly-agile robots - Boston Dynamics
 - Humanoid robot - Tesla 2022



Today's Robotics

- 2025 - 2026: Humanoid robot boom in China
 - Commercial & Mass Produced: Sub-\$16,000 price disruption by Unitree (G1/R1) and AgiBot scales hardware globally.
 - Deployed in EV Factories: Real-world industrial pilots by UBTECH (Walker S) in NIO, BYD, and Foxconn assembly plants.
 - Embodied AI Driven: Transitioning from classical rigid control to multimodal Vision-Language-Action (VLA) models.
 - Strategic Scale: **China accounts for >50% of global humanoid patents and an estimated 10,000+ units manufactured by 2025/2026.**



Unitree G1: \$16k
bipedal platform

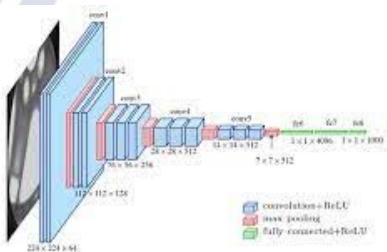


AgiBot: Dual-arm
embodied AI

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Now is a good time to study and work on Embodied AI



Artificial Intelligence

Recent breakthroughs in machine learning, language and computer vision: CLIP, ChatGPT. Expect a good Large Vision-Language Model in recent years



Computing Power

Your smartphone is millions of times more powerful than all of NASA's combined computing in 1969.



¥17332.00



xArm 5/6/7轴机械臂 UFACTORY 众为创造

¥ 44999.00 已售: 0件

Robot Hardware

More reliable and affordable robot hardware. Boston Dynamics' spot costs \$75,000 Unitree has \$10,000 ones



3D Simulators

More open-source, easy-to-use 3D simulators.
*CARLA
*NVIDIA Isaac Sim

Now is a good time to study and work on Embodied AI

- Covid-19 makes us realize the importance of Embodied AI that limits human interactions.



By Brianna Wessling | February 6, 2023

ABB uses robots to automate COVID antibody testing

The system, created by ABB and UTMB, can increase the number of neutralizing antibody tests performed from 15 a day to over 1,000 daily.

By Steve Crowe | January 8, 2021

Ava Robotics launches UV disinfection robot

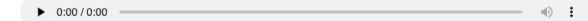
Ava Robotics, the iRobot spinout known for its telepresence robots, is expanding its product portfolio with the introduction of an ultraviolet (UV) disinfection robot for corporate offices, warehouses, and other work spaces. The new robot is an evolution of a research project Ava Robotics conducted with the Massachusetts Institute of Technology and the Greater Boston...



9 robotics projects focusing on COVID-19

By Steve Crowe | September 24, 2021

Listen to this article



Voced by Amazon Polly



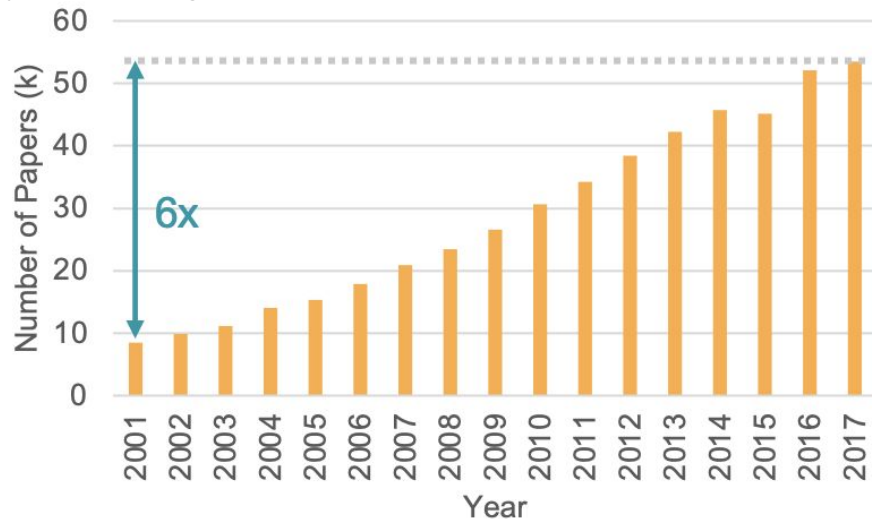
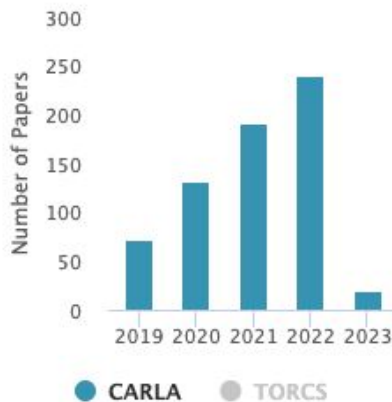
<https://www.therobotreport.com/tag/coronavirus/>

Now is a good time to study and work on Embodied AI

- The research community is rapidly growing
 - Publications at top conferences are on the rise!
 - ICRA, CoRL, RSS, IROS, etc.
 - Papers using 3D simulator is rapidly increasing

Why conferences are better than journals (most of the time)?
Because the research happens fast...

Usage 



Number of paper on robot learning

Embodied AI is challenging!

From DARPA Robotics Challenge 2015

A 3-stage competition to develop robot assistants. Tasks involved:

1. Robot drives through an obstacle course to destination and exit car
2. Enters a building through door
3. Turns a valve
4. Cuts a hole in a wall using a tool
5. Climb stairs to finish

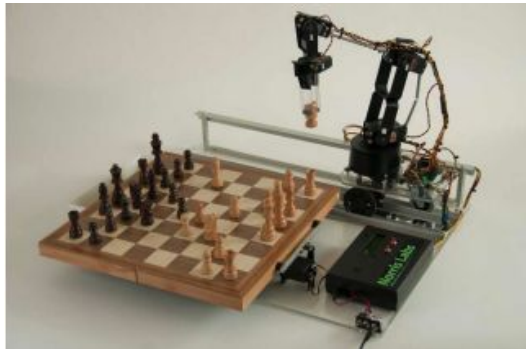


Only 3 teams successfully finished all the tasks

Embodied AI is challenging!

You may hear of “The Moravec's paradox” in the robotics literature

- The Moravec's paradox (1988):
 - “it is comparatively easy to make computers exhibit adult level performance on intelligence tests or playing checkers, and difficult or impossible to give them the skills of a one-year-old when it comes to perception and mobility”



Still true in some sense nowadays. You can ask ChatGPT to write a convincing story but it could not do your laundry

Embodied AI is challenging!

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 - “it is comparatively easy to make computers exhibit adult level performance on intelligence tests or playing checkers, and difficult or impossible to give them the skills of a one-year-old when it comes to perception and mobility”

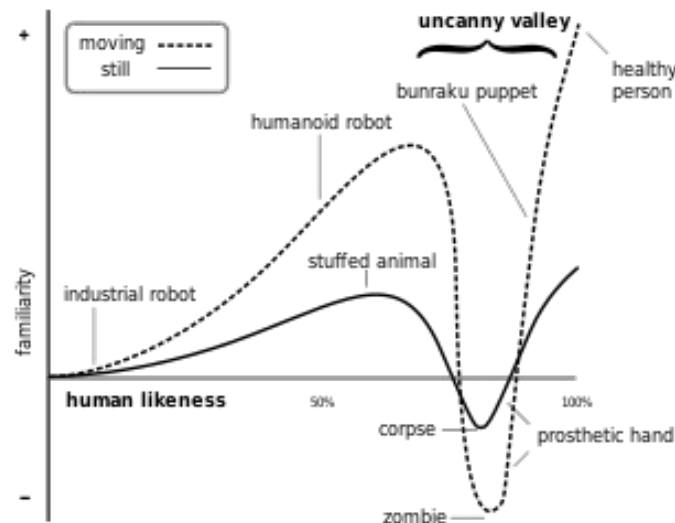


Boston Dynamics has the most advanced highly-agile robots. But they are not selling it (and at an affordable price) at any time soon.

Embodied AI is challenging!

You may also hear of The uncanny valley (恐怖谷理论)

- The uncanny valley (恐怖谷理论):
 - “...a hypothesized relation between an object's degree of resemblance to a human being and the emotional response to the object”
 - It means that if you want to build humanoid robot that looks like a human, it needs to be very good otherwise people will hate it
 - At some point we will need to consider the impact of human-computer interaction



As the robots become more like human, our familiarity with it may suddenly drops to a negative level

Agenda

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- History of Robotics
- Why study Embodied AI now?
- **Course content overview**
- Logistics

Key Components in Embodied AI



Perception

Seeing and understanding of surroundings



Decision Making

Motion planning and control of robot behaviors



Robot Systems

Robot embodiment.
3D simulations.
Sim2real

Key Components in Embodied AI



Perception

Seeing and understanding of surroundings



Decision Making

Motion planning and control of robot behaviors

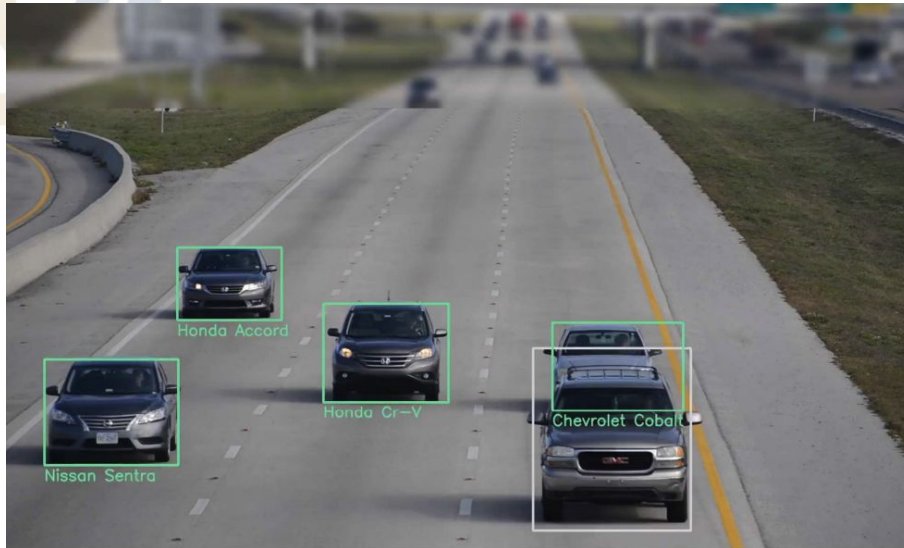


Robot Systems

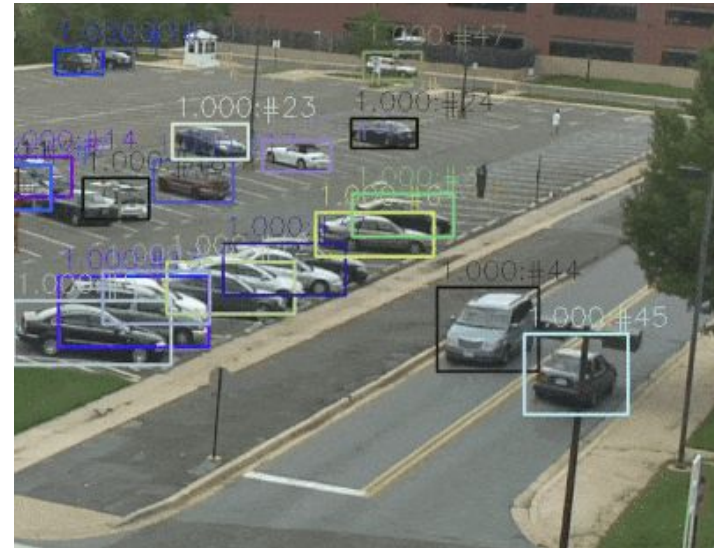
Robot embodiment.
3D simulations.
Sim2real

Embodied AI - Perception

- Typical tasks include Object detection & tracking



Drawing bounding boxes of different classes of objects

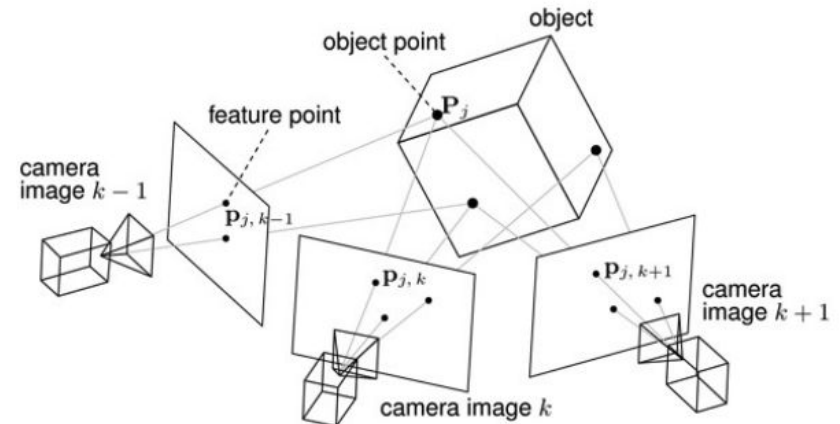
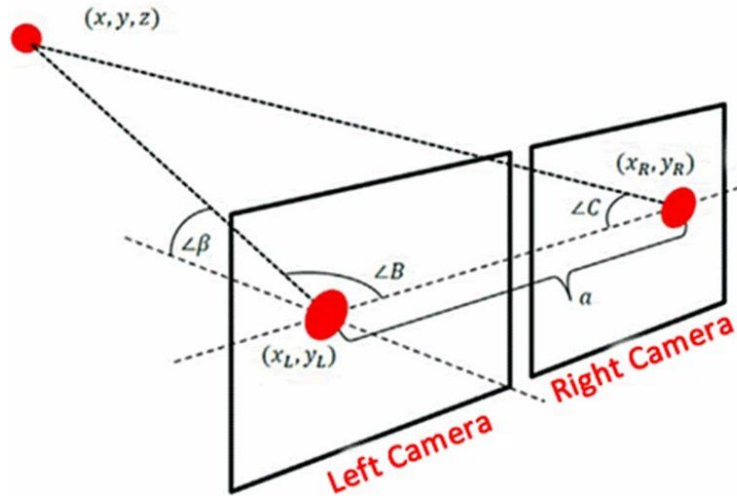


Tracking: Associate an ID for each object

Embodied AI - Perception

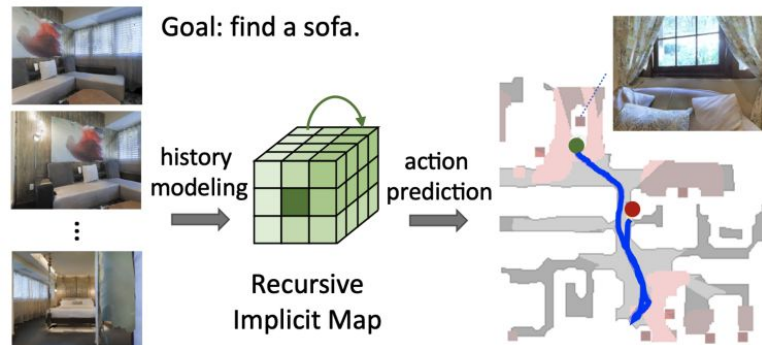
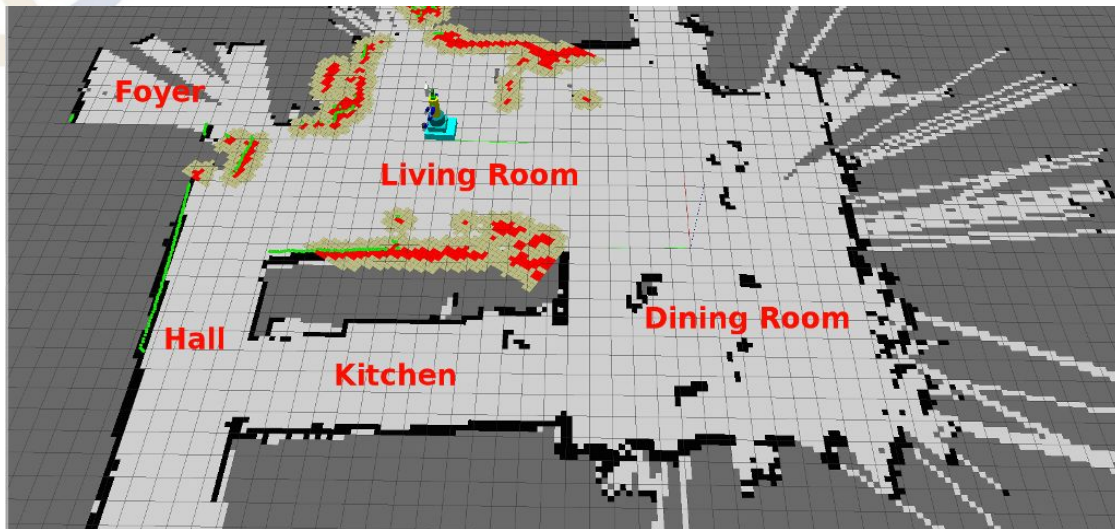
- Stereo Vision

- Can be used to estimate depth using geometrics of multiple cameras (will cover in this course)
- And used for 3D reconstruction



Embodied AI - Perception

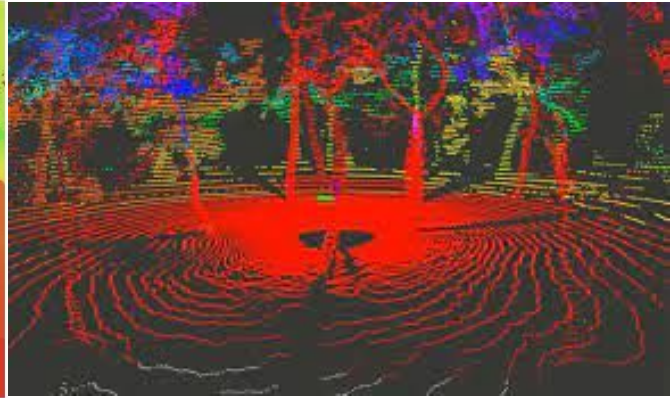
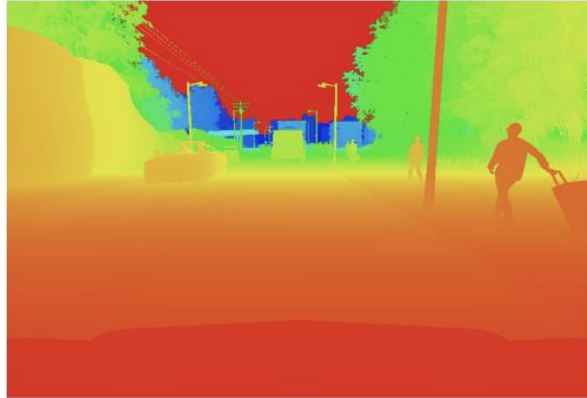
- Localization and mapping
 - Important for indoor robots
 - Could be done explicitly like in SLAM or implicitly



*Object Goal Navigation with Recursive Implicit Maps. IROS 2023

Embodied AI - Perception

- Multimodal Sensor Fusion
 - Combine information from RGB, Depth, and point clouds (LIDAR)



Embodied AI - Perception

- 3D Visual Grounding - SeeGround (CVPR 2025)
 - Grounding: given language query, locate the object in the 3D scene



Key Components in Embodied AI



Perception

Seeing and understanding of surroundings



Decision Making

Motion planning and control of robot behaviors

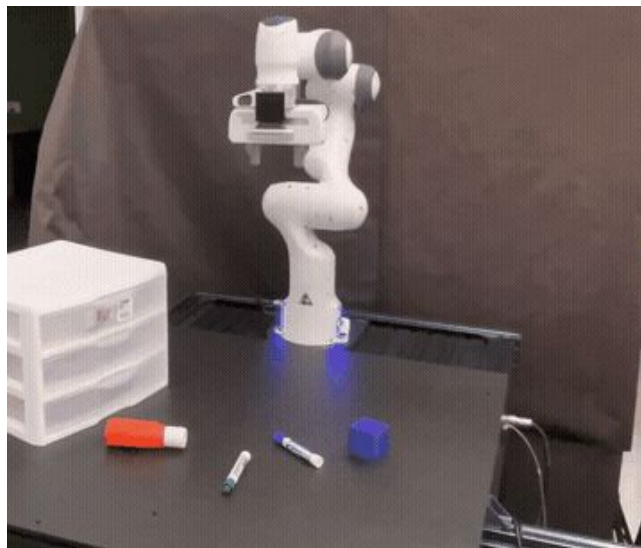


Robot Systems

Robot embodiment.
3D simulations.
Sim2real

Embodied AI - Decision Making

- Motion Planning and Control
 - Model outputs future motion plans for the robot to execute



*RVT: Robotic View Transformer for 3D Object Manipulation. CoRL 2023



*Dit4Dit, 2026

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Robot Systems

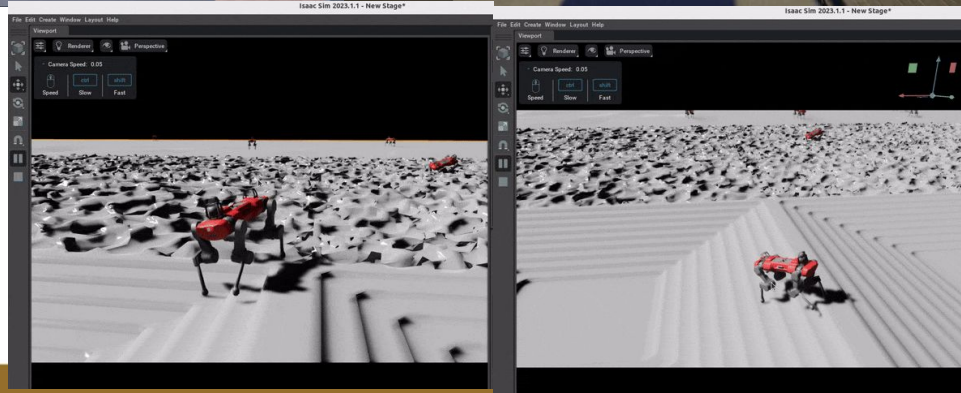
Robot embodiment.
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Embodied AI - Robot Systems

- 3D Simulation is a big part of robotics research



* CARLA simulator
(based on UE4):
<https://carla.org/>
<https://leaderboard.carla.org/>



* Nvidia Isaac Sim:
<https://developer.nvidia.com/isaac-sim>

Embodied AI - Robot Systems

3D simulator
can be very
realistic

This is from
CARLA -
Specialized
for
self-driving
task

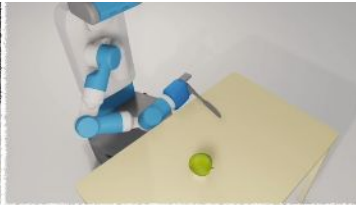


Embodied AI - Robot Systems

- 3D Simulation is a big part of robotics research



Wiping Table with Cloth



Slicing Apple with Knife

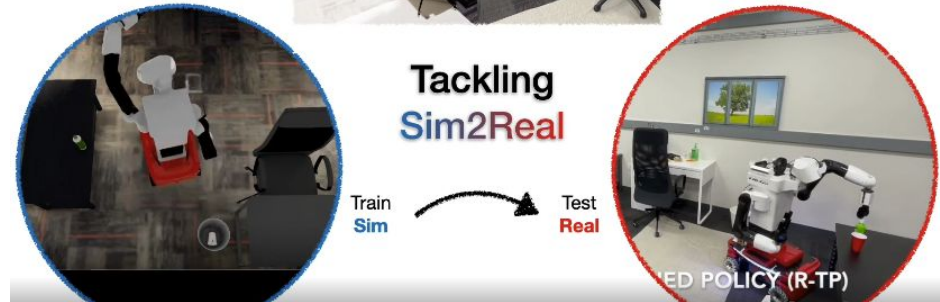
Complex Robot Behaviors

Ecological Environments

Setting Dinner Table



Cleaning Bathroom

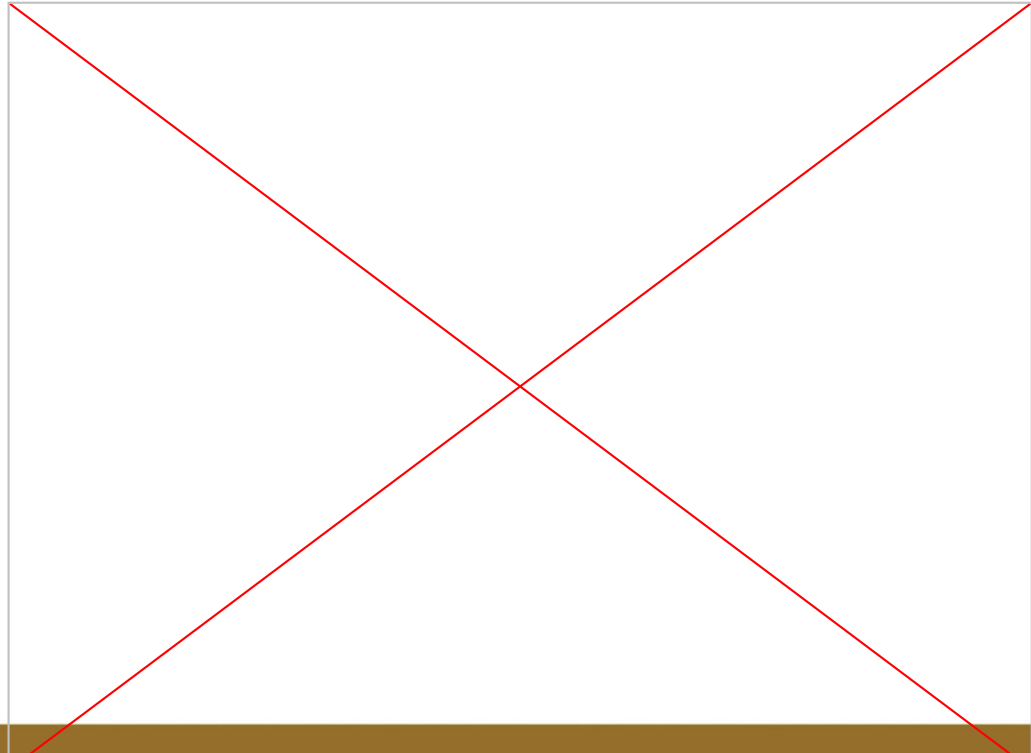


* BEHAVIOR-1K (Based on **Isaac Sim**): <https://behavior.stanford.edu/behavior-1k>

Embodied AI - Robot Systems

- 3D Simulation is a big part of robotics research

This is from BEHAVIOR-1k -
Specialized for everyday task
robots



Key Components in Embodied AI



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Course Plan

- Machine Perception
 - Geometric computer vision
 - Visual Recognition
 - Localization and Mapping
- Robot Systems
 - ROS
 - Simulation
- Decision Making
 - Planning and Control
 - State estimation, Kinematics and dynamics
 - **Embodied Navigation**
 - **Markov Decision Process, Reinforcement Learning, etc.**
 - **Embodied Interaction and Manipulation**
 - **VLA, etc.**

Course Objectives

- Describe key principles, concepts, models and techniques in Embodied AI
 - Understand technical challenges and recent research progress
- Implement algorithms introduced in class to solve small real-world problems
 - Familiarity with Embodied AI systems like ROS and 3D simulators
- Develop novel methods for Embodied AI applications
 - Navigation
 - Manipulation

Course Objectives

- Develop novel methods for Embodied AI applications
 - By the end of the course you can build your navigation robot dog that can avoid humans!

Falcon in diverse scenarios (Social-HM3D)

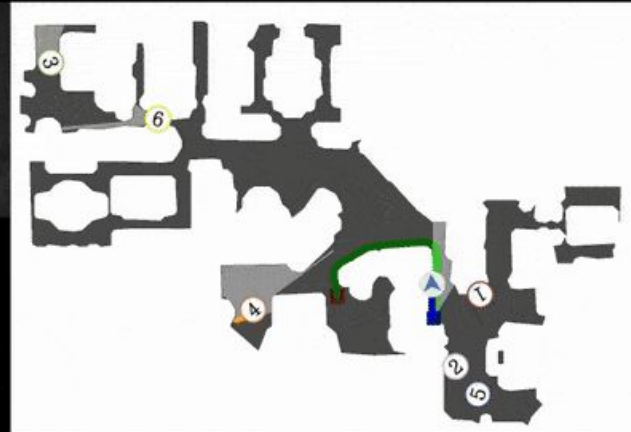
Third-person View



Depth



Top Down Map



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Logistics: Part 1

- Course website

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Embodied AI

AIAA 4220, Fall 2026

Instructors: [Junwei Liang](#)



香港科技大学 (广州)
THE HONG KONG
UNIVERSITY OF SCIENCE AND
TECHNOLOGY (GUANGZHOU)

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	In-Class Quizzes	

Logistics: Part 1

- Canvas
 - <https://hkust-gz.instructure.com/courses/3918>
 - Course announcement will be posted here
 - Use the discussion section for questions and discussions
 - **We will use Canvas for in-class quizzes**
- Compute resources
 - We will provide you each with a ¥200 credit for university HPC for project 2
 - Courtesy of AI Thrust
 - We will distribute the account/password after 9/14
 - Aliyun 阿里云 : <https://university.aliyun.com/> free ¥300 RMB credit
 - * There will be 5 simulation projects you can choose from for Project 3
 - Most of them does not require a GPU

Logistics - Computation Resources from Aliyun

- Aliyun has provided our university with free or discounted computation resources
 - <https://university.aliyun.com/>
 - 300 RMB voucher
 - 70% off for orders up to 5000 RMB/year
 - You can directly apply with your hkust-gz email

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Logistics: Part 2

- Course aims to give you more hands-on experience
 - Means coding and experimenting
- Prerequisites
 - Python programming, Basic calculus, linear algebra, probability theory
 - You should really be using Ubuntu (not Windows) - Most codes/tools are based on that
- Language of choice: Python
- Toolkit of choice: PyTorch / Habitat Sim / Mujoco / Isaac Sim
 - You are welcome to use other languages/toolkits, but the TAs will not be able to help
- Participation
 - Interrupting (by raising a hand) to ask your question is strongly encouraged
 - Asking questions later on Canvas (or in class) is also great
 - Interaction improves learning (both in-class and at my office hours)
- Office hours
 - Rm 304, E4
 - Wednesday 10:00AM - 11:30AM (Please email me first to make an appointment)

Grading

- A range (A+, A, A-): normally requires an overall course grade of at least 80 AND Ranked in the top 35%;
 - A+: normally requires an overall course grade of at least 95 and exceptional performance across the course AND Ranked in the top 10%
- B range (B+, B, B-): normally requires an overall course grade of at least 70.
- C range (C+, C, C-): normally requires an overall course grade of at least 60.
- D: normally requires an overall course grade of at least 50.
- F: normally applies to an overall course grade below 50.

The above grading rules may be subject to modifications depending on the university regulations and the actual distributions of the course grades.

Grading - Overview

- Course evaluation overview
 - (25%) 6 in-class quizzes, multiple-choice questionnaires (**Grading: best 5 out of 6**)
 - (25%) Project 1: In-class Paper Presentation - Single-person work
 - (25%) Project 2: Egocentric Object Detection - Single-person work
 - (25%) Project 3: Simulation Project - Group of 2-3 people
 - We do not take attendance! But you should be here for the in-class quiz and your paper presentation

Grading - Details

- In-class Quizzes (25%)
 - Designed to complement the lectures and making sure you understand the key knowledge about the course
 - 6 individual assignment (so you need to complete this yourselves and no LLM allowed)
 - Grading based on your best 5 out of 6 (so you can miss one and still get full 100%)
 - **Will be posted on Canvas Quizzes section**
 - Schedule
 - 10:10 - 10:20 am.
 - Test 1: Sep 21
 - Test 2: Oct 12
 - Test 3: Oct 26
 - Test 4: Nov 9
 - Test 5: Nov 23
 - Test 6: Nov 30

Grading - Details

- Project 1: Paper Presentation (25%)
 - You will select 1 paper from our list of Embodied AI research to present. This should show that you understand the paper, and you could present it to students who do not know the paper well. **You have 10 minutes to present and 5 minutes to handle at least one question (from TA/peers/teacher)**
 - Paper list: [link](#)
 - Full schedule
 - P1 presentation order release on 9/21
 - P1 presentation #1-8 from 10/12 to 11/30
 - (5 students per lecture) (starting at 10:10 am.)
 - This is an individual assignment
 - Project 1 is worth 25% credit
 - The grading is based on both the presentation (50%) and the slides (50%), by the teacher and TAs. Please follow the slide template here: [link](#)
 - You are required to submit the presentation slides after the presentation

Grading - Details

- Project 2: Machine Perception Homework (25%)
 - You will do an Ego-view object detection homework
 - This is an individual assignment
 - You will submit to a Kaggle leaderboard.
 - **Grading based on model absolute score and project report**
 - Detailed grading rubrics: [link](#)
 - Three absolute score thresholds will be published
 - S: full credit / A: 80% / B: 70%
 - Tentative schedule:
 - P2 release on 9/21
 - Kaggle submission due on 10/26
 - Report due on 10/30

Grading - Details

- Project 3: Simulation Project (25%) - Done in groups of 2-3 people
 - You can choose from 5 tracks:
 - MiniGrid: Sparse-reward navigation
 - BabyAI: Language-guided imitation learning
 - MuJoCo: Robust continuous control
 - ManiSkill: Tabletop manipulation
 - Habitat: Collision-aware navigation
 - Task: Reproduce a baseline, introduce one motivated modification, and evaluate it through a fair, controlled comparison. Analyze both successes and failures.
 - Deliverables: 15-minute presentation (12 min + 3 min Q&A) and report
 - Project grading: 50% presentation + 50% report. Grading rewards sound experiments and clear analysis; outperforming the baseline is not required.
 - Tentative schedule:
 - P3 handbook release on 10/19
 - Form a group by 11/2
 - Presentation on 12/7; Report and slides due on 12/20

Grading - Details

- Late Policy

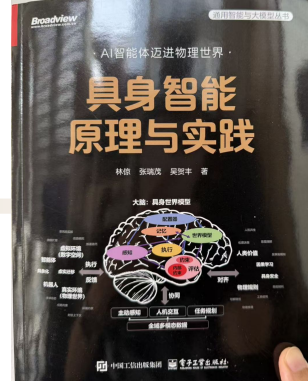
- Homework assignments and course project results are worth full credit on the due date. Unless granted an extension in advance, it is worth at most 75% credit for the next 48 hours, at most 50% credit after that. If you need an extension, please ask for it as soon as the need for it is known. Extensions that are requested promptly can be granted more liberally. You must turn in all assignments.
- **There will be no make-up quizzes**

Grading - Details

- Use of LLM/VLMs (ChatGPT, Gemini, etc.)
 - We encourage you to use ChatGPT to solve the homework assignment problems (because likely in the future everyone will use it for coding). However, you will need to declare using ChatGPT/LLMs in the homework README, as well as the prompts you use
 - **You are not allowed to use LLM for in-class test!!!**

More Course Resources

- Textbooks
 - Machine Learning: A Probabilistic Perspective - Kevin P. Murphy
 - Multiple View Geometry in Computer Vision - Richard Hartley and Andrew Zisserman
 - Introduction to Autonomous Mobile Robots - Illah Reza Nourbakhsh and Roland Siegwart
 - Chinese textbook: 具身智能原理与实践 by my professor friends
- Workshops and Competitions
 - Precognition @CVPR [\[link\]](#)
 - Workshop on Autonomous Driving @CVPR [\[link\]](#)
 - CARLA Autonomous Driving Challenge @NeurIPS [\[link\]](#)
- Reading Materials
 - Nice talks
 - [A Path Towards Embodied AI - Yann LeCun](#) [\[link\]](#)
 - Robotics Professor Answers Robot Questions From Twitter - Henny Admoni [\[link\]](#)
 - Other Open Courses
 - Stanford AA 274a
 - CMU 16-311, 11-851
 - Washington CSE 478
 - Texas CS 391R
- **More online tutorial videos/Important knowledge about Embodied AI by Prof. Junwei Liang**
 - **[Tencent Docs]** Embodied AI学习指引 for UG
 - <https://docs.qq.com/sheet/DREN4UUpyaVBvbGdL?tab=qgffmc>



Your ToDOs

- Check course webpage, Canvas
 - <https://hkust-aiaa4220.github.io/fall2026/index.html>
- Review linear algebra, calculus and probability
- Review python coding, numpy, Ubuntu system
- Checkout 3D simulators and play around
 - <https://carla.org/>
 - <https://behavior.stanford.edu/>
 - <https://developer.nvidia.com/isaac-sim>
- Start checking out the paper to read
 - Embodied AI paper list: [link](#)
- Start finding group members to pair with others in course projects!



Thank you for deciding to take this journey of Embodied AI with us

Hope you can learn something and get a good grade!

Any questions?